Qn			Answers	Marks
2	(a)		Newton's Law of Gravitation states that the force of attraction between any two **point** masses is **directly** proportional to the product of their masses and **inversely** proportional to the square of their separation.	B1
	(b)	(i)	$F_{Earth} = \frac{GMm}{r^2} = \frac{6.67\times10^{-11}(5.98\times10^{24})6200}{(1.50\times10^{9})^2}$	M1
			$F_{Sun} = \frac{GMm}{r^2} = \frac{6.67\times10^{-11}(1.99\times10^{30})6200}{(1.50\times10^{9}+1.496\times10^{11})^2}$	M0
			Resultant force = gravitational force by earth + gravitational force by sun = 1.10 + 36.0	M1
			= 37.1 N	A1
	(b)	(ii)	$F_{net} = mr\omega^2$ $37.1 = 6200(\underbrace{1.50\times10^{9}+1.496\times10^{11}}_{1.511\times10^{11}})\omega^2$ [Allow ECF from (b)(i)]	M1
			$\omega = 1.99\times10^{-7}$ rad s^{-1}	A1
	(b)	(iii)	The satellite remains at a **constant** distance of 1.50×10^9 m from the Earth, on the line joining the centres of the Earth and Sun. (The satellite, Earth and the Sun are collinear).	A1

	(c)		Gravitational force provides the centripetal force. $\frac{GMm}{r^2} = m\frac{v^2}{r}$ $v = \sqrt{\frac{GM}{r}} = \sqrt{GM}\ r^{-\frac{1}{2}}$	M1
			$k = \sqrt{GM}$	A1
			$n = -0.5$	A1
	(d)		When the speed v of the satellite decreases, the centripetal force $\frac{mv^2}{r}$ required for a circular orbit of the same radius r decreases.	B1
			The gravitational force $\frac{GMm}{r^2}$ is initially unchanged and is therefore greater than the centripetal force $\frac{mv^2}{r}$. As a result, there is an unbalanced inward force acting on the satellite in the direction towards the centre of the Earth which cause the satellite to move towards the Earth (distance between the satellite and the Earth decreases).	B1